

Khoa Thai Dang Tran

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PROFESSIONAL SUMMARY

Software engineer and MS Computer Science student at the University of the Pacific, with 2+ years of professional experience across full-stack development, databases, and AI-powered applications. Strong foundation in data structures & algorithms, database design, and automated testing/CI pipelines. Seeking a Software Engineering role to contribute to impactful products.

EDUCATION

University of the Pacific, Stockton, CA

January 2025 - Present

MS Computer Science | GPA: 3.76 - Expected graduation: December 2026

University of Science, VNU - HCM, Vietnam

2018 - 2022

BS Information System | Graduation: 2022

EXPERIENCE

Software Engineer

November 2022 - December 2024

UNICCS Co., Ltd

- Architected and delivered full-stack features for retail management systems used by clients in Japan, building backend services in Java Spring Boot and responsive frontends in JavaScript and Vue.js.
- Improved SQL query efficiency and system performance through query tuning, indexing, and schema design on PostgreSQL and DB2, supporting high-volume transaction processing.
- Deployed and maintained services on AWS, collaborating with a cross-functional team to keep the platform stable and production-ready; automated deployment, build, and monitoring workflows with Shell scripting.
- **Tech:** JavaScript, HTML, CSS, PostgreSQL, DB2, Shell, Vue.js, SQL, Java Spring Boot, AWS

Software Engineer Intern

December 2021 - June 2022

FPT Telecom

- Built RESTful APIs with Node.js and Express to power POP management dashboards for real-time telecom monitoring.
- Developed interactive dashboards in React and Next.js, integrating Highcharts to visualize data across multiple POP stations.
- Designed and integrated MongoDB schemas to store and query telecom monitoring data efficiently.
- **Tech:** React.js, Next.js, Express, Node.js, MongoDB, Highcharts

SKILLS

- **Languages:** JavaScript, Python, Java, TypeScript, C++, C#, HTML, CSS, SQL, TSQL
- **Frameworks:** React.js, Next.js, Vue.js, Node.js, Express, Spring Boot, Bootstrap, Tailwind CSS
- **Databases:** PostgreSQL, MySQL, MongoDB, DB2, SQL Server, Oracle
- **Testing & CI/CD:** Vitest, Playwright, GitHub Actions
- **Tools:** Git, GitHub, SVN, Slack, AWS, Shell, Highcharts, Jupyter Notebook, Claude Code (AI-assisted development)
- **Concepts:** Object-Oriented Programming/Design, Data Structures & Algorithms, RESTful APIs, Database Design, Full-Stack Development, Agile/Scrum, LLM Integration & Prompt Engineering, Agentic/Tool-Use AI Systems

PROJECTS

CineVibes - Personal Project

- Built a full-stack movie discussion platform with spoiler-aware, category-based comment threads, trending rankings, and CineBot, an in-app AI chat assistant (Groq-hosted LLM) for movie recommendations and trivia, grounded in the app's own MongoDB catalog via a search tool.
- Designed the test strategy and automation framework: unit/integration tests (Vitest), end-to-end tests (Playwright), and a GitHub Actions pipeline running the full test suite, lint, and build on every push/PR.
- **Technologies:** React, Node.js, Express, MongoDB, Vite, JWT Auth, Groq, Vitest, Playwright, GitHub Actions.

ML-Assisted Natural Language to SQL Query System - Personal Project

- Built a CSV-agnostic natural language to SQL system: users upload any CSV and ask questions in plain English. Pairs a BERT intent classifier (fine-tuned on 1,250 examples across 14 domains) with a self-built, schema-aware rule-based SQL generator - no external LLM API calls.
- **Technologies:** Python, BERT (bert-base-uncased), PyTorch, Streamlit, SQLite, Pandas, Matplotlib, Seaborn.

ExpenseTracker - Personal Project

- Developed a client-only expense tracker (logging/filtering, budgets, dashboard analytics, CSV/JSON/PDF export) built end-to-end in close collaboration with Claude Code: compared three parallel implementations of the export feature, caught and fixed a CSV-injection security bug the comparison surfaced, and verified every change with lint, a strict-mode TypeScript build, and the Vitest suite before merging.
- **Technologies:** Next.js, React, TypeScript, Tailwind CSS, Recharts, jsPDF, Vitest.